

Class Character**TODO 1:**

```
public abstract sealed class Character permits Warrior, Mage {
```

TODO 2:

```
protected String name;
```

```
protected int healthPoints;
```

TODO 3:

```
public Character(String name, int healthPoints) {  
    this.name = name;  
    this.healthPoints = healthPoints;  
}
```

TODO 4:

```
@Override  
public String toString() {  
    return name + " has " + healthPoints + " health points."  
}
```

TODO 5:

```
@Override  
public boolean equals(Object obj) {  
    if (null == obj) return false;  
    if (this == obj) return true;  
  
    if (obj instanceof Character other)  
        return name.equals(other.name) && healthPoints ==  
other.healthPoints;  
  
    return false;  
}
```

TODO 6:

```
public abstract void attack();
```

Class Warrior

TODO 7:

```
non-sealed class Warrior extends Character {
```

TODO 8:

```
public Warrior(String name, int healthPoints, int armor) {  
  
    super(name, healthPoints);  
    this.armor = armor;  
}
```

TODO 9:

```
public int getArmor() {  
    return armor;  
}
```

```
public void setArmor(int armor) {  
    this.armor = armor;  
}
```

TODO 10:

```
@Override  
public void attack() {  
    System.out.println(name + " swings a sword!");  
}
```

Class Mage

TODO 11:

```
final class Mage extends Character {
```

TODO 12:

```
@Override  
public void attack() {  
    System.out.println(name + " casts a spell!");  
}
```

TODO 13:

```
@Override  
public String toString() {  
    return super.toString() + " Mana: " + mana;  
}
```

Class Guild

TODO 14:

```
public static final int MAX_MEMBERS = 10;
```

TODO 15:

```
public Guild() {  
  
    members = new Character[MAX_MEMBERS];  
}
```

TODO 16:

```
public boolean searchMember(Character character) {  
    for (int i = 0; i < nbrMembers; i++) {  
        if (members[i].equals(character))  
            return true;  
    }  
    return false;  
}
```

TODO 17:

```
public void addMember(Character member) {  
  
    if (nbrMembers >= MAX_MEMBERS || searchMember(member)) {  
        return;  
    }  
  
    members[nbrMembers] = member;  
    nbrMembers++;  
}
```

TODO 18:

```
public void levelUp() {  
  
    if (nbrMembers < 2) return;  
  
    int mages = 0;  
    int warriors = 0;  
    for (int i = 0; i < nbrMembers; i++) {  
        if (members[i] instanceof Mage)  
            mages++;  
        else  
            warriors++;  
    }  
  
    if (mages < 2 || warriors < 4) {  
        System.out.println("Not enough members!");  
        return;  
    }  
    level++;  
  
    for (int i = 0; i < nbrMembers; i++) {  
        if(members[i] instanceof Mage mage)  
            mage.setMana(mage.getMana() + 100);  
        else if(members[i] instanceof Warrior warrior)  
            warrior.setArmor(warrior.getArmor() + 50);  
    }  
}
```